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ORIGINAL RESEARCH

The perils of approximate ontology

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Abstract
The mathematical centerpiece of many physical theories is a Lagrangian. So let's imagine that there's some Lagrangian we trust. Should that induce us to endorse an ontology? If so, what ontology, and how is it related to our trustworthy Lagrangian? I'll examine these questions in the context of quantum field theoretic Lagrangians. When these Lagrangians are understood as "merely effective," a variety of approximations figure in the physics they frame. So do distinctive grounds for trusting those Lagrangians, grounds recent literature has adduced in support of a novel variety of scientific realism known as *Effective Realism*. This essay attempts to undermine those grounds, and to do so without presupposing extensive prior knowledge of quantum field theories.

Keywords Approximation · Scientific realism · Effective field theories · Quantum theories · Ontology

1 Introduction

The mathematical centerpiece of many physical theories is a Lagrangian. In classical mechanics, the Lagrangian and the principle of least action generate the Euler–Lagrange equations, which determine what histories are possible for a system. That space of histories frames a classical theory's canonical quantization. In contexts both classical and quantum, Lagrangians guide empirical predictions—even when we don't know how to solve the equations of motion those Lagrangians generate. Often—and especially, as in the case of Lagrangians underlying interacting quantum field theories (QFTs), when solution spaces exceed our mathematical grasp—the guidance is mediated by approximation techniques. Mediated though it is, the guidance nevertheless emanates from a Lagrangian that constitutes the theoretical core of the approach in question.

So let's imagine that there's some Lagrangian we trust. Should that induce us to endorse an ontology? If so, what ontology, and how is it related to our trustwor-

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thy Lagrangian? I'll examine these questions in the context of QFT Lagrangians. When these Lagrangians are understood as "merely effective," a variety of approximations figure in the physics they frame. So do distinctive grounds for trusting those Lagrangians, grounds recent literature (e.g., Fraser, 2016; 2018; 2020a; Miller, 2017; Williams, 2019) has adduced in support of a novel variety of scientific realism branded *Effective Realism* by its proponents. I'll try here to undermine those grounds. I'll suggest that an effective Lagrangian, one we trust for the distinctive reasons celebrated by Effective Realists, isn't merely a starting point from which approximation techniques depart. There's a sense in which it, itself, is an approximation. And, I'll argue, the approximateness of effective Lagrangians complicates, and even confounds, the project of articulating an ontology for QFT worth endorsing. Complicated and confounded as well is the aspiration to strike a realist stance toward that ontology. This unsettles both the contents of, and the reasons to endorse, an effective theory's "approximate ontology."

Although the views are presented in different vocabularies and motivated by analyses of different grains, I take the position I defend here to be a close relative of Ruiz de Olano et al., (2022)'s *Content Determination Thesis*. Before proceeding to the business at hand, it's worth sketching how I take the views to be related to one another, as well as to broader themes in the philosophy of physics, and beyond. Both de Olano et al. (2022) and I resist a conventional account of the *content* of a successful physical theory T, that is, of what T represents the world as like. This conventional account attributes T a *theoretical core*.[1] For the interacting QFTs considered here, that theoretical core consists in a Lagrangian $\mathcal{L}$ (or Hamiltonian or action or whatever else can play the possibility-determining role described in the first paragraph—although I'll speak here of Lagrangians, I take the morals to generalize to other sorts of theoretical core). The conventional account continues: this theoretical core $\mathcal{L}$ encodes T's physical and empirical content. I'll use "ontology" to refer to this content.[2] Please note that "ontology" so understood comprehends not only what a theory says there is, but also its account of *how* what there is *behaves*. To make the realism debate interesting, I'll assume that T has content that outruns its empirical predictions—T asserts that there are gluons which mediate the strong force, say. What distinguishes realists about T from anti-realists, on this way of setting things up, is that the former endorse (at least some of) T's transempirical ontology. Realists believe (at least some of) what T says about what lies beneath the phenomena, and take T's success to license this belief. Antirealists do not.

[1] Roughly correlate to Ruiz de Olano et al's notion of an "idealized model." The dialectic framing their treatment foregrounds questions about modeling and idealizations; the dialectic framing mine foregrounds questions about realism and representation. While I think it's possible and useful to supply a thoroughgoing translation manual between our two settings, doing so here would take me too far afield.

[2] According to a notion of "interpretation" I think is useful, one thing an interpretation of T might do is specify its ontology. I'm framing things here in terms of "ontology" to avoid having to explain how the notion of interpretation I think is useful differs from other notions of interpretation available in the effective field theory literature, e.g., Williams' (2019) "standard interpretation" and Dougherty's (2023) "standard account of interpretation." (Concerning the latter: I take Dougherty's radical effective realist to be a relative of the unpristine interpreter—where there's a family squabble over whether there's still room for scientific realism at the dinner table).

Further realist-friendly thoughts are allied with, but not entailed by, the conventional account. One thought is that, even if there are different ontologies one might associate with the theoretical core $\mathcal{L}$, one of these will be, as it were, the "winning ontology," the account of $\mathcal{L}$'s physical content that both makes the best sense of T's success *and* successfully represents the physical world (see North, 2022, Ch. 1–2, for one version of this thought developed). Another is that grasping this winning ontology constitutes *understanding* T. A third is what Ruetsche (2011) terms "the ideal of pristine interpretation," restyled here as "the ideal of pristine ontology." To equip T with a pristine ontology is to insulate the content of that ontology from considerations (other than overarching ones of good taste) exogenous to T's nomic core $\mathcal{L}$. It's to extract an ontology from $\mathcal{L}$ by appeal only to perfectly general considerations, considerations of mathematics and/or metaphysics and/or methodology, say. The ideal of pristine ontology is that $\mathcal{L}$ on its lonesome harbors an account of what, according to T, there is; and that aspiring ontologists should strive to exercise the acuity necessary to articulate that account. The contrast class, unpristine ontologies, invoke lowly brutish contingencies when characterizing T's ontology, where these auxiliary contingencies aren't (contra the "surrogative reasoning" picture Ruiz de Olano et al., de Olano et al. (2022) discuss) merely crutches or shortcuts to an ontology wholly immanent in $\mathcal{L}$, but substantive and ineliminable contributors to T's physical content. These contingencies might include initial and boundary conditions, pragmatic considerations such as intended applications and calculational resources, or the current shape of aspirations toward future theories. In this vernacular, Ruiz de Olano et al. (2022)'s *Content Determination Thesis* is: when it comes to interacting QFTs, approximation techniques coupled to, but not determined by, $\mathcal{L}$ have an ineliminable hand in articulating T's (thereby unpristine!) ontology. They support the thesis through the careful analysis of historical examples, some of which surface briefly in Sect. 4.3.

Put metaphorically, in the case of pristine ontology, $\mathcal{L}$ is ontologically legible to the (sufficiently astute) naked eye. For unpristine ontologies, $\mathcal{L}$ becomes ontologically legible only when viewed through purpose-ground spectacles afforded by specific approximation techniques or other considerations too muddied by brutish contingencies to qualify as pristine. Nothing in principle prevents an unpristine ontology from being a winning ontology. But recognizing the possibility, that $\mathcal{L}$ isn't ontologically legible on its own, should prime us to take seriously another possibility, one that threatens to undermine the realist position sketched above. Different unpristine ways of reading ontologies into $\mathcal{L}$ could make sense of different aspects of T's successes—with no single reading emerging as a *winning* ontology, an ontology realists can give abductive grounds to accept. And different ways of reading ontologies into $\mathcal{L}$ can look different, in a way that matters to the realism debate, in hindsight. From the perspective of future physics, different ways of reading $\mathcal{L}$ can be more, or less, mistaken. The Pessimistic Meta-Induction would leverage this circumstance into an argument against realism: the history of science is littered with ontologies that were once winning but are now discarded. Realists can disrupt that argument if they can give reasons to expect the persistence, through future episodes of theory change, of the ontology they identify, now, as the winning ontology—if they can give reasons to think that

whatever privileges an ontology as winning also renders it *resilient*.[3] By contrast, an anti-realist who takes different unpristine ontologies to manifest different virtues in different circumstances need not identify any of those ontologies as resilient. Such an anti-realist countenances a more open future for physics than the realist.

What, in terms of these general themes, I'm going to try to argue here is that, when $\mathcal{L}$ is an effective Lagrangian we trust for distinctive grounds adduced by Effective Realists, there are significant impediment to reading a resilient ontology off of $\mathcal{L}$ with the naked eye. I proceed as follows: Sect. 2 gives lightning reviews of QFT Lagrangians, particle physics strategies-in-practice for reading ontologies off of them, and prima facie challenges to reading resilient ontologies into them. Introducing the effective theory idea, Sect. 3 reviews distinctive resources Effective Realists develop for reading an effective Lagrangian's ontology, and for trusting that ontology's resilience. Section 4 contends that, rather than dispelling challenges to the ontological legibility of QFT Lagrangians, the Effective Realism maneuver amplifies them, and amplifies them in a way that undercuts the resilience of the ontologies in contention. Section 5 will, very, very briefly, advocate an attitude toward the contents of QFTs that enables us not only to live with but also to celebrate the ontological illegibility of effective Lagrangians. I aim throughout to make my discussion accessible to readers who are interested in interactions between physics and metaphysics, but who don't specialize in the history and foundations of QFT.

2 QFT Lagrangians and their (dis)contents

There's a "physicsy" way of thinking about the ontology of QFTs, including those making up the terrifyingly empirically successful Standard Model of contemporary particle physics. These *folk ontological strategies* take the Standard Model to describe a variety of entities, entities it characterizes by attributing them nomically significant traits, including mass, charge, and spin. These traits' nomic significance derives from their role in the Standard Model's account of how those entities behave—the traits help dictate not only how the entities (would) evolve in isolation but also how they interact with one another, and even what kind of statistics their aggregates obey. So, for instance, the folk ontology recognizes an *electron neutrino*, which it attributes mass < 2.2 eV, charge 0, spin $\frac{1}{2}$ (so collectives of electron neutrinos obey Fermi–Dirac statistics); the electron neutrino is left-handed and has weak isospin $\frac{1}{2}$; thanks to these additional attributes, it interacts in a parity-violating way through the weak force. And so on. It is standard practice to call these entities "particles," and I will follow suit. But here I'll use the label without prejudice to finer-grained questions of particle metaphysics (what Chakravarttry (2017) might term "Big-M metaphysics"), such as those pursued in Falkenburg (2007).

[3] I am grateful to Dave Baker for inducing me to recognize the question of whether resilience decouples from winning.

2.1 Lagrangians

The Lagrangian for a *field theory*, not unsurprising, involves *fields*, where a field is a function that assigns every point of space(time) a "value." A homely example is an ocean wave, which assigns a real number—the wave's height—to every point on the ocean floor. The ocean wave field takes numbers (aka scalars) as values; it's a scalar field. But fields of other sorts map points of spacetime to "values" with more internal structure than scalars have: for instance values that are pairs or triples or …of real or complex numbers, or $n \times n$ matrices, or….

When the equations of motion determined by a classical Lagrangian are exactly soluble, there are procedures for constructing explicitly from this classical solution space the full mathematical apparatus of a quantum theory; the quantum theory explicitly constructed is the *canonical quantization* of the classical theory identified with the original Lagrangian (see Folland, 2008, Ch. 5 for details). Most interesting QFTs—indeed, every known interacting QFT in four spacetime dimensions, including the QFTs making up the Standard Model—correspond to more obstreperous Lagrangians. We don't know how to solve their equations of motion. Instead of building a quantum theory from a precisely-characterized classical solution space, we resort to a toolkit of approximation techniques incorporating characteristically quantum mechanical principles to wrest, from the Lagrangian, quantities of experimental interest ("scattering cross sections," "amplitudes," "correlation functions"). Feynman diagrams guide the most widely-taught method for calculating these approximations. As Sect. 2.3 will rehearse, these calculations have some philosophically provocative aspects.

The Lagrangian underlying a QFT will be a function of some collection of fields and their spacetime derivatives. Here's a Lagrangian involving only one field $\phi(x)$, a real scalar field, a map from points x of spacetime to real numbers. (To make things easier to read, $\phi(x)$ will often be abbreviated ϕ in what follows; other fields will follow suit and drop their spacetime arguments for display purposes.)

$$\mathcal{L}_{\phi_4} = \frac{1}{2}(\partial\phi)^2 - \frac{1}{2}m^2\phi^2 - \lambda\frac{\phi^4}{4!}$$

Through a slight abuse of terminology I'll cop to presently, I'll call monomonials of ϕ and its spacetime derivatives, such as $(\partial\phi)^2$, ϕ^2, and ϕ^4, the *interactions* appearing in $\mathcal{L}_{\phi_4}$; the coefficients modifying the interactions are *couplings*.

Once we've settled on a list $\{O_i\}$ of interactions that might appear in a Lagrangian, we can express a generic Lagrangian as the sum

$$\mathcal{L} = \sum_i g_i O_i$$

where the g_i are $\mathcal{L}$'s couplings. It'll be good practice for what lies ahead to conceive of Lagrangians like $\mathcal{L}_{\phi_4}$ as residents of a infinite-dimensional *space* $\mathcal{G}$ of Lagrangians; interactions that could appear in a Lagrangian correspond to axes for the space. Specifying a Lagrangian, its couplings g_i also specify a point in $\mathcal{G}$.

2.2 Folk ontological guidelines

QFT is also particle physics. The folk ontology associates particles with fields appearing in QFT Lagrangians. Although the association between particles and field excitations deserves scrutiny, I'll accept it here as a working assumption, for the sake of focusing on other questions about the ontology of QFT.

Thus: $\mathcal{L}_{\phi_4}$ describes one particle species, the particle associated with the ϕ field. Call this species a *ϕ-particle*. The Lagrangian

$$\mathcal{L}_{QED} = \bar{\psi}(i\gamma^{\mu}\partial_{\mu} - m_e)\psi - \frac{1}{4}(F_{\mu\nu})^2 - e\bar{\psi}\gamma^{\mu}A_{\mu}\psi$$

is more interesting than $\mathcal{L}_{\phi_4}$. While the latter is a tractable(ish) toy model, $\mathcal{L}_{QED}$ underwrites an empirically successful interacting QFT, Quantum ElectroDynamics, which describes how photons mediate an electromagnetic force between (electrically) charged particles. According to the folk ontology, $\mathcal{L}_{QED}$ describes two species of particle, one (which has particle and antiparticle subspecies) associated with the $\psi/\bar{\psi}$ field—call this species *$\psi/\bar{\psi}$ particles*—and one associated with the A_{μ} field, which everyone calls a *photon*. The γ^{μ} may look like a field, but it's really just a gadget for smoothing the mathematical joints in interactions involving $\psi/\bar{\psi}$ fields. As for the $F_{\mu\nu}$, it's a shorthand defined in terms of A_{μ} and its derivatives.

The folk ontology takes *how* fields appear in Lagrangians to code nomically significant traits of the particle species associated with those fields. The coding works in several ways.

2.2.1 "Type" of field

One is via the "type" of field associated with the particle species. Fields are of "different type" when they reside in spaces bearing different concrete representations of key symmetry groups. So, for example, the real scalar field ϕ, the vector field A_{μ}, and the spinor field $\psi/\bar{\psi}$ are different types of fields because they live in spaces—the spaces of real numbers, vectors, and spinors, respectively—on which different representations of the Lorentz group act. Fields in different spaces transform differently under the Lorentz group.[4]

A key symmetry group is the Poincaré group of spacetime symmetries, consisting of translations, rotations and boosts (which relate reference frames moving at a constant velocity with respect to one another). Wigner showed how to label representations of the Poincaré group with quantities we can understand as mass and spin (for more careful statements, see Williams, 2023, Sect. 3.1; Talagrand, 2022, Ch. 8). This suggests to the folk ontology that the *type of field* on which a *representation of the Poincaré group* acts codes the mass and spin of the particle species associated with the field. Parity—mirror image reflection—is a discrete spacetime symmetry. In

[4] Scalar fields transform trivially under the Lorentz-group. Vector- and matrix- valued fields—e.g., the four-component Higgs field—can be scalar fields in this sense. The Higgs transforms non-trivially under other symmetry groups—underscoring that the same field can provide different "types" of representation of different symmetry groups.

some representations of the Poincaré group extended to include parity transformations, those transformations act trivially; in other representations of the extended Poincaré group, they do not. Fields that live in representations of the first sort don't come in left-handed and right-handed varieties; the associated particles lack chirality. Fields that live in representations of the second sort are right- or left- handed, as are their associated particles.

Similar rules of thumb hold for internal symmetries, symmetries that act on a field's "internal structure" (if it has one!). Quark fields, for instance, live in a three-dimensional space on which a representation of QCD's internal $SU(3)$ symmetry acts. Think of a basis for this space supplied by a trio of states labelled *blue*, *green*, and *red*. You've just imagined quark fields as a type of field that supports the ascription of a range of color properties to their associated particles. Color is a significant trait because particles with that trait interact through QCD's strong force.

2.2.2 Couplings

Another way the folk ontology takes a Lagrangian to code physically significant particle traits is through the couplings modifying terms in that Lagrangian. The bilinear $\frac{1}{2}(\partial\phi)^2$ in $\mathcal{L}_{\phi_4}$ resembles lowbrow kinetic energy by depending on the gradient squared of a configuration. Physicists are wont to call such terms *kinetic terms*. Along with the *mass term* $-\frac{1}{2}m^2\phi^2$, this kinetic term describes the *free* (that is, non-interacting) behavior of the ϕ particle. The coupling coefficient m in $\mathcal{L}_{\phi_4}$'s mass term alerts the folk ontology that the ϕ particle has mass m. Kinetic term and mass terms involve fields to second order; terms third-order or higher in their participating fields are *interaction terms*.[5] A splendid example is $\mathcal{L}_{\phi_4}$'s term $\lambda\frac{\phi^4}{4!}$, a *self-interaction* term. (In the absence of other fields, ϕ has only itself to interact with!) Recalling that Lagrangians subtract potential energy from kinetic energy, we can interpret this term to encode the potential energy $U(\phi)$ of the field's self-interaction. Thus the coupling λ gives the interaction strength of this self-interaction. Similarly, the coupling e in the QED Lagrangian interaction term $e\bar{\psi}\gamma^\mu\psi A_\mu$ tells the folk ontology that electrons have a charge e that governs how strongly they interact electromagnetically.

Thus the folk ontologist reads $\mathcal{L}_{\phi_4}$ to describes a single particle species, the ϕ-particle, which is a *real scalar particle*, because ϕ is a real scalar field; that ϕ is a scalar field, rather than a more structured field, tells her that the ϕ particle has no spin, chirality, or electric charge, and that the ϕ particle is its own antiparticle; $\mathcal{L}_{\phi_4}$'s $\frac{1}{2}m^2\phi^2$ term tells us her that the ϕ particle has mass m; the coupling λ in $\mathcal{L}_{\phi_4}$'s self-interaction term tells her how strongly the ϕ particle interacts with itself. The absence from $\mathcal{L}_{\phi_4}$ of further interaction terms tells her that the ϕ particle remains aloof from further interactions, e.g., electromagnetic. The lesson already drawn, that real scalar particles lack a charge through which they'd interact electromagnetically, corroborates this suggestion. Likewise the folk ontologist takes $\mathcal{L}_{QED}$ to describe

[5] So the terminological abuse I commit by calling every monomial of fields and their spacetime derivatives that might appear in a Lagrangian an "interaction" is that the monomials $(\partial\phi)^2$ and ϕ^2 aren't interactions. My excuse is that another standard umbrella term for the mononomials, "operator," invites confusion with Hilbert space operators. And, as already noted, if a Lagrangian $\mathcal{L}$ has interactions, we don't know how to build from its solution space a Hilbert space in which to set its quantization.

$\psi/\bar{\psi}$ particles, which are *spinor particles*, because ψ and $\bar{\psi}$ are spinor-valued fields. That ψ /$\bar{\psi}$ are spinor-valued fields tells her that the $\psi/\bar{\psi}$ particle has an antiparticle, and that these particles and antiparticles have spin $\frac{1}{2}$, carry an electric charge, and are "handed" or chiral (that is, they transform non-trivially under parity flips). Particles and antiparticles have opposite charges and chiralities. $\mathcal{L}_{QED}$'s kinetic term + mass term $\bar{\psi}(x)(i\gamma^{\mu}\partial_{\mu} - m_e)\psi(x)$ tells the folk ontologist that they (particle and antiparticles both) have mass m_e. The folk ontologist reads $\mathcal{L}_{QED}$ to also describe an A_{μ} particle, the photon, which is a vector particle, because A_{μ} is a vector field. It follows that the photon has spin 1 (so it's a boson), no electric charge and no chirality. This makes the photon its own antiparticle. Although a kinetic term involving the A_{μ} field is hiding in the $(F_{\mu\nu})^2$ interaction, there is no mass term. This tells the folk ontologist that the photon is massless. $\mathcal{L}_{QED}$'s interaction term $e\bar{\psi}\gamma^{\mu}\psi A_{\mu}$ describes electromagnetic interaction between $\psi/\bar{\psi}$ particles mediated by photons. The coupling e is the electric charge of the $\psi/\bar{\psi}$ particle.

These examples illustrate how a trusty Lagrangian discloses to the folk ontology not only what particles species there are—species corresponding to fields appearing in the Lagrangian—but also what nomically significant properties distinguish those species from one another—properties like mass, charge, spin, and color, coded either by the the types of fields populating the Lagrangian or by its couplings. A trusty Lagrangian encodes what physicists often call a "particle spectrum," and what I'll call a *particle ledger*: a roster of particle species (the ledger's headings) and their nomically significant traits (entries under each ledger heading).

2.3 Perturbative approximation and the folk ontology

It's worth noting how the topic of this special issue—approximation—plays a quiet but crucial role in the folk ontology. What lends aid and comfort to the idea that QFTs describe particles is the fact that explicitly-realized QFTs have a Fock Space structure (see Williams, 2023, Sects. 2.1–2.2 for an introduction). By "explicitly realized QFTs", I mean QFTs we can successfully subject to the techniques of canonical quantization mentioned at the outset of Sect. 2.1. The resulting Hilbert space of possible QFT states is a Fock space. A Fock space is a sturdy scaffolding for a particle interpretation: a Fock space supports decidedly particulate quantum observables (for instance, number operators that count particles); Fock space states, including a vacuum or no-particle state, can be understood in terms of particle contents limned by these observables. This might prime us to expect that, associating a particle spectrum with a Lagrangian, the folk ontology makes implicit appeal to a Fock space structure.

When the Lagrangian is an interacting QFT Lagrangian, the appeal is mediated by an approximation. An approximation is required because we don't known how to canonically quantize an interacting Lagrangian, and hence cannot construct from it a Fock space scaffolding supporting a particle notion. We can, however, canonically quantize *free* Lagrangians, that is, Lagrangians containing only kinetic and mass terms (e.g., $\mathcal{L} = \frac{1}{2}(\partial\phi)^2 - \frac{1}{2}m^2\phi^2$). Now consider an interacting Lagrangian $\mathcal{L}_{int}$ whose interaction couplings are small. Differing from a free theory $\mathcal{L}_0$ only by the presence of measly perturbations corresponding to its interactions, $\mathcal{L}_{int}$ lies in the *perturbative*

regime. That is, $\mathcal{L}_{int}$ is almost the free Lagrangian $\mathcal{L}_0$! It is "standard practice" (Ruiz de Olano et al., 2022, p. 85) in particle physics to appeal to the Fock space scaffolding we know how to construct from $\mathcal{L}_0$ to characterize the particle spectrum of the interacting theory based on $\mathcal{L}_{int}$. Bedeviling the practice is Haag's theorem, which will make another appearance in Sect. 4.1, and which suggests that non-trivial interactions can't be defined in the particle-notion- supporting Fock space constructed from the free theory.

This is our first whiff of "approximate ontology": $\mathcal{L}_{int}$ is "approximately" $\mathcal{L}_0$, in the mathematically tractable sense that the couplings modifying $\mathcal{L}_{int}$'s interaction terms are small numbers. Does it follow that $\mathcal{L}_{int}$'s *ontology* is approximately $\mathcal{L}_0$'s? To what do we appeal when issuing such judgements of ontological similarity? Realists owe us an answer, insofar as a notion of ontological approximation is arguably implicit in realist responses to the Pessimistic Meta-induction. Where $\mathcal{L}_{now}$ and $\mathcal{L}_{future}$ are once and future Lagrangians, a winning ontology for $\mathcal{L}_{now}$ will be resilient insofar as it's similar to $\mathcal{L}_{future}$'s winning ontology. Section 4 will revisit the matter of ontological approximation.

More straightforward, at least initially, than the notion of ontological approximation, is the idea that we can, on behalf of an interacting Lagrangian $\mathcal{L}_{int}$, *perturbatively approximate* experimental quantities. Suppose $\mathcal{L}_{int}$ adds a small perturbation to a free Lagrangian $\mathcal{L}_0$. Schematically

$$\mathcal{L}_{int} = \mathcal{L}_0 + g\mathcal{O}$$

where the coupling g is small. To perturbatively approximate an empirical quantity $\mathcal{M}$—a scattering amplitude, say—on behalf of $\mathcal{L}_{int}$, is to calculate $\mathcal{M}$ via a series of corrections, in ever higher orders of the coupling, to the (usually pretty boring) explicitly calculable value $\mathcal{M}_0$ predicted by the free theory.

$$\mathcal{M} = \mathcal{M}_0 + g\mathcal{M}_1 + g^2\mathcal{M}_2 + \cdots$$

The terms $\mathcal{M}_1$, $\mathcal{M}_2$, etc., are perturbative corrections that take the interaction into account. It's the job of perturbation theory to guide the calculation of these corrections. Given that g is small, we can dare to hope that if we can work out the first few perturbative corrections, we'll have a tolerably accurate—good enough for empirical purposes—numerical approximation to $\mathcal{M}$.

For interacting QFTs, Feynman diagrams are a heuristic for keeping track of perturbative corrections: each Feynman diagram corresponds to a mathematical expression we solve to calculate such a correction. Notoriously, starting at second order, Feynman diagrams instruct us to calculate by constructing and evaluating integrals that *diverge*—go to infinity. These divergences render the perturbative approximation empirically senseless. Undaunted, mid-twentieth century physicists developed a toolkit of *perturbative renormalization* techniques for taming the divergences. Through the careful pursuit of these methods, integrals contributing at each order to perturbative corrections can be rendered finite. The rendering relies on empirical inputs, in the form of a finite number of magnitudes measured in particle accelerator experiments conducted at a reference energy scale known as the *renormalization scale*. A Lagrangian

is *perturbatively renormalizable—renormalizable*, for short—if techniques of perturbative renormalization, coupled with a finite number of empirical inputs, issue finite corrections at each order.

For present purposes, the details of perturbative renormalization methods don't matter.[6] What does matter are two things.

- *First*, renormalizable interacting Lagrangians are rare in the following sense. Recall the space $\mathcal{G}$ of Lagrangians involving a scalar field. In principle, each Lagrangian in this space could depend non-trivially on infinitely many interactions $\mathcal{O}_i$. The select minority of renormalizable Lagrangians depend only on a particular, and finite, set of *renormalizable* interactions. A Lagrangian is renormalizable when all of its interactions are. Although $\mathcal{L}_{\phi_4}$ is renormalizable, adding additional interaction terms (proportional to ϕ^6, say) would spoil that. In general, given a set of fields, there's only a finite set of *renormalizable interactions* you can build from them. Renormalizable Lagrangians are rare in the sense that they reside in a finite-dimensional subspace—the subspace spanned by renormalizable interactions—of an infinite-dimensional space of possible Lagrangians.

- *Second*, the theories making up the Standard Model of particle physics are renormalizable—and the predictions they yield when subject to techniques of perturbative renormalization are not only finite but also absurdly accurate. This prompts the question of "why renormalizable theories play such an important role in physics" (Schwartz, 2014, p. 443)—why the physics of the actual world happens to be governed by one of the small set of Lagrangians whose divergences we can tame at all orders using techniques of perturbative renormalization. Section 3 will outline an answer to this question which is at the same time an inspiration for Effective Realism.

A Lagrangian whose small interaction couplings render it apt for perturbative approximation is also a Lagrangian that might (modulo concerns about the very idea of "ontological approximation"!) avail itself of the particle spectrum of the corresponding free theory. de Olano et al. (2022) illustrate their Content Determination thesis by considering *strongly*-coupled physical theories, theories for which the foregoing ontological strategy breaks down—and showing how the approximation techniques applied to such theories shape the ontologies associated with them.

The next subsection suggests ontological difficulties—impediments to associating a resilient ontology with a Lagrangian—that could arise, even in the perturbative regime. Section 3 introduces a version of Effective Realism that promises to suppress such difficulties. Subsequent sections argue that, rather than resolving difficulties of this sort, Effective Realism exacerbates them.

[6] Conceptual introductions to perturbative renormalization include Fraser (2020a), Williams (2023, Sects. 4.1–4.2) (both aimed at philosophers of science) and Zee (2010), Part III (aimed at physics students). There's no impediment to calculating finite predictions on behalf of a non-perturbatively renormalizable Lagrangian, provided we're willing to truncate the perturbative corrections at a finite order.

2.4 Ontological illegibility

A particle ledger is a schematic ontology. The virtue of *ontological legibility* graces a QFT Lagrangian when it's straightforward to read a particle ledger off of that Lagrangian. The realist aspires to do more than read an ontology off a Lagrangian. She wants reasons to trust what she reads, including reasons to consider her ontology secure against disruptions from future physics.

This section offers a prima facie challenge to the resilient ontological legibility of a QFT Lagrangian. It does not resolve the question of how much this challenge should trouble the realist. But it raises that question. An apparent virtue of Effective Realism is the solace it offers to realists who are troubled by the question.

Consider a Lagrangian written in terms of two real fields ϕ_1 and ϕ_2.

$$\mathcal{L} = \frac{1}{2}(\partial\phi_1)^2 - \frac{1}{2}m^2\phi_1^2 + \frac{1}{2}(\partial\phi_2)^2 - \frac{1}{2}m^2\phi_2^2 - g(\phi_1^2 + \phi_2^2)^2 \tag{1}$$

That very same Lagrangian could also be written in terms of a vector-valued field $\Psi = \begin{pmatrix} \phi_1 \\ \phi_2 \end{pmatrix}$. Or the Lagrangian (1) could be written in terms of a complex field $\psi = \phi_1 + i\phi_2$ that has a $U(1)$ symmetry we can ally with a conserved "charge" q implementing a representation $\theta \rightarrow exp(-iq\theta)$ of $U(1)$ on the space $\mathbb{C}$ of field states (see Brading & Brown, 2003 for more about how Noether's theorem connects symmetries and properties). I could keep going, but we have enough ways to write the Lagrangian (1) to illustrate "changes of ontological basis." To review, those ways include (a) in terms of two real scalar fields ϕ_1 and ϕ_2; (b) in terms of vector-valued fields $\begin{pmatrix} \phi_1 \\ \phi_2 \end{pmatrix} \in \mathbb{R}^2$; and (c) in terms of complex scalar fields $\psi = \phi_1 + i\phi_2 \in \mathbb{C}$.

Applying folk ontological strategies to the Lagrangian (1) written in terms of (a) eventuates in a particle ledger featuring two species of real scalar particle, each (in an eerie coincidence) of mass m. Applying folk ontological strategies to the Lagrangian (1) written in terms of (b) eventuates in a particle ledger featuring one species of mass m vector particle. Applying folk ontological strategies to the Lagrangian (1) written in terms of (c) eventuates in a particle ledger featuring a mass m complex scalar particle species and its antiparticle. This particle ledger boasts an entry for charge (q for the particle, $-q$ for the antiparticle). These three particle ledgers correspond to three (at least superficially) distinct ontologies.

Supposing that there are significant ontological differences between these particle ledgers, the failure of Lagrangian (1) to dictate *which* particle ledger to read off of it compromises its ontological legibility. But the supposition is a stretch. Metaphysicians (e.g., those anthologized in Chalmers et al., 2009) who can hear high-pitched differences between very similar-sounding existence claims might have it in them to distinguish between the particle ledgers. (Indeed, it would be just like a metaphysician to distinguish invidiously between the ledgers on the grounds that ledgers (b) and (c), doing away with the "mass coincidence" besetting ledger (a), are simpler than (a) is.) But philosophers of physics tend to take a theory's content to reside in its mathematical structure. (For a recent, and sophisticated, expression of the tendency,

see Wallace, 2022.) To their eyes, ledgers (a)–(c) look like notational variants on one another, differences in the convention used to describe a common ontology.

But even their attitude might change if it's disclosed that (1) contains a typo: that the fields ϕ_1 and ϕ_2 should appear with *different* masses m_1 and m_2. Then there is something to choose between the ledgers—the Lagrangian can no longer be written in a form that supports ledger (b) or (c). Or suppose it were disclosed that the Lagrangian has further terms in which the charge q acts as a coupling. Then particle ledger (c) starts to look not only different from but also superior to the others: (c) builds a charge trait into the nature of its particles by associating them with representations sustaining a charge notion.

It might seem that these outrageously toy examples undermine the "change in ontological basis" challenge, by making it vivid that it's not the original Lagrangian, but different modifications to it, that invite different readings. But here's another way to spin the examples, in two cycles. The first spin cycle: The examples illustrate that the couplings and interactions appearing in a Lagrangian matter to how we "ontologize it."[7] Second spin cycle: as physics progresses, we might *modify* the Lagrangians we favor at present by amending their couplings and their interactions. The ontologies we read off the Lagrangians we favor at present are resilient if they're sufficiently stable under those modifications—if the modifications are appropriately understood as telling us a bit more about the particles we already recognize and/or as characterizing *additional* particles in a way that extends our ontology without significantly altering its present contents.

Dare realists about current theory hope to read resilient ontologies off of QFT Lagrangians? To temper that hope, I'll sketch a expurgated history of the neutrino, introduced (in a 1930 letter that opens with the salutation, "Dear Radioactive Ladies and Gentlemen") by Pauli to explain the continuous energy spectrum observed in decay processes. Fermi's 1932 "4 Fermi" theory of beta decay (elaborated in Sect. 4.2) uses a Lagrangian in which the neutrino appears as a mass 0 neutral fermion field that interacts directly with three other fermions (e.g., a neutron, the proton into which it transitions, and the electron that's a decay product) in beta decay. Figure 1 gives a Feynman diagram for this 4 Fermi interaction. Later theories of the weak force refine this picture: featuring additional fields and additional couplings, their Lagrangians portray beta decay as mediated by the exchange of a massive gauge boson, the $W-$boson (see Fig. 2).

[7] For a potential extreme example, in 2 spacetime dimensions, of the sensitivity of ontology to couplings, consider the sine-Gordon Lagrangian, which describes a $1+1$ dimensional massless scalar field ϕ subject to a potential that depends on a real parameter β, and the Thirring model, which describes a $1+1$ dimensional massive fermionic field whose interactions depend on a coupling g. The weakly coupled (small values of β) sine-Gordon Lagrangian shares a functional form with the strongly coupled (large values of g) Thirring Lagrangian. Despite agreeing in functional form, one Lagrangian evokes a particle ledger featuring massless bosons, the other a particle ledger featuring massive fermions. Sydney Coleman comments, "One of the most striking features of the results established here is that a theory which is 'obviously' a theory of fermions is equivalent to a theory which is 'obviously' a theory of bosons" (1975, 2094). What exactly to make of this—Coleman favors a "*nuclear democracy*" according to which "a single theory has two equally valid descriptions in terms of Lagrangian field theory: the massive Thirring model and the quantum sine-Gordon equation" (Coleman 1975, 2086)—is not a question this footnote will settle. For some starts on it see McKenzie (2017) and Castellani (2017).

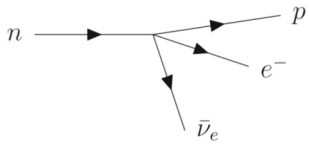

Fig. 1 Beta decay, according to the 4-Fermi Lagrangian

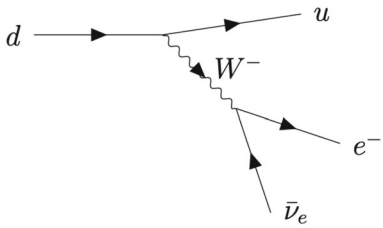

Fig. 2 Beta decay, according to the Electro-Weak Theory. To relate to Fig. 1, note that changing a down quark d in a neutron n to a up quark u changes n into a proton p

In 1954, nuclear physicists are exercised by the τ-θ puzzle: τ and θ, mesons with the same mass and lifetime, decay into products with different parities. So either they're different particles that just happen to be dopplegangers, or weak decays don't conserve parity. In 1957, Chien-Shiung Wu conducts elegant experiments confirming that weak interactions violate parity conservation. The Lagrangian governing weak interactions is accordingly modified so that it's no longer invariant under interchange of right- and left-handed fields. Now it describes neutrinos all of which are left-handed.

It wouldn't be a stretch to read the story thus far as one in which we're modifying our neutrino ontology at the edges (we weren't wrong to think they were fermions; we've just also learned that they're left-handed fermions), and embedding it in a more encompassing ontology (we weren't wrong to think there were neutrinos; we've just also learned that there are $W-$ bosons too). That is, it can be read as one throughout which the neutrino ontologies we read off our favored Lagrangians are resilient. Only the story continues. Starting in the 1960s, solar neutrino detectors (enormous vats of cleaning fluid or water or heavy water housed deep underground and surrounded by photomultipliers tuned to be triggered by Cherenkov radiation) register responses that don't align with theoretical predictions. These data, collected not in new generations of particle accelerators achieving previously unheralded energies but in abandoned mines, are a (unresolved) problem for the Standard Model. For reasons this history omits (but which Burgess and Moore (2006), Ch. 10, relate), this solar neutrino problem could be addressed through a mechanism (neutrino flavor oscillation) that requires neutrinos to be massive. It is not, however, straightforward to make neutrinos massive. If they're garden-variety fermions governed by the Dirac equation, then they can have a mass only if they come in both left- and right-handed chiralities. But where are the right-handed neutrinos? One possibility is that they're out there, just supermassive and "sterile"—that is, they don't interact with quarks, leptons, or gauge bosons—so it's no wonder we haven't noticed them. It is a mystery why neutrinos of different chiralities have such different masses, though. A second possibility is that neutrinos aren't Dirac fermions after all, but another kind of fermion field, called a Majorana field, that can be left-handed and massive. But this possibility poses mysteries of its own: for instance, it's consistent with interactions in which leptons change "flavors"—a τ-neutrino becomes an electron neutrino, say—but we've not observed such interactions.

A third possibility is that neutrinos come in all four—right- and left- handed, Dirac and Majorana—fermion varieties. And a fourth possibility is latent in each of the other three. It's that we'll eventually settle on an accommodation of anomalous underground mine evidence conveyed by a theory so unlike the 4-Fermi theory, and so unlike the present Standard Model, that neutrinos don't appear in it at all. Even if none of the first three possibilities strikes you as a threat to the resilience of ontologies read off of the Standard Model, I would suggest that the fourth one should.

And, if you're a realist, I have a further outlandish suggestion to make. It's that ontologies for the Standard Model could be lurking in each of the first three possibilities sketched above. Whereas realists should be reluctant to read the Standard Model "at face value" to posit massless neutrinos, lest their commitments come to grief on underground mine data, ontologies keyed to the first three possibilities sketched above have a fighting chance to survive the tribunal of such data. The outlandish suggestion foregrounds the sort of ontological illegibility that matters in the context of Pessimistic Meta-induction. To resist that argument, *realists need to read into the Lagrangians we have* ***now*** *the ontologies we'd attribute them in* ***hindsight***. This section challenges the realist to make a case that they'll be able to pull this off. (There's a possible (admittedly loose) connection to the ideal of pristine ontology here: which ontology we judge most likely to be resilient interacts with how we understand, and prosecute, projects of theory development—and that in turn could interact with the sort of contingencies (accidents of professional training, or available conceptual or material technologies) that adulterate an ontology.)

This section has mounted a challenge: it could be difficult to read resilient ontologies off of QFT Lagrangians. Here are a few reactions to the challenge, ordered from least to most supportive of the variety of realism sketched in Sect. 1.

- Admit that resilient ontologies for QFT Lagrangians might not be legible to the naked eye. For example, to associate a particle ledger (that isn't bankrupt by the solar neutrino puzzle) with the Lagrangian of the Standard Model is to cast one's lot with a particular strategy for making sense of neutrino mass. Likewise for other QFT Lagrangians: they could invite different ontologies in (and, according to the Content Determination Thesis, sometimes *through*) different circumstances of approach or application. Admit as well that, not knowing the future of physics, we have no principled handle on the question of which candidate ontology (if any) will be resilient. We have no response to the Pessimistic Meta-Induction.
- Walk back what we expect of ontology. If we walk it back so far that the different ontologies attributable to a Lagrangian $\mathcal{L}$ are deemed to agree, we've restored $\mathcal{L}$'s ontological legibility, essentially by collapsing the different texts we might see in it to a single narrative. An interesting question here is: how long a walk will this take? If the uncertainty about the ontological basis through which to read a Lagrangian were the only complication, a walk this long will suffice: an ontology doesn't tell us what particles species there are, or what their nomically significant traits are. Rather, it tells us *how many degrees of freedom* it takes to specify the states of the set of fields (however parameterized) appearing in $\mathcal{L}$. A further question is whether we've now walked back so far that we've left interesting ontology behind. "7 dimensional," is, after all, a decidedly disappointing response

to the question,"What's your ontology?"—and there's room to wonder whether a commitment to 7 degrees of freedom on its own constitutes a realism in any interesting sense. When uncertainty about the future of physics distorts ontological focus, it's harder to characterize how to walk expectations back—but perhaps even easier to doubt that, once they're walked back, the commitments that meet them constitute a realism in any interesting sense.

- When challenged to assign a resilient ontology to a Lagrangian $\mathcal{L}$, produce reasons to privilege a commital, substantive ontology, the kind of thing that's correct or incorrect according to whether there really are particles species of the sort it posits—and also produce reasons to expect that ontology to be resilient in the face of future theory change. This is the reaction most consistent with realism: the privileged ontology is $\mathcal{L}$'s "winning ontology," the ontology a realist endorses, an endorsement the realist supports by appeal to $\mathcal{L}$'s success, and defends against the Pessimistic Metainduction by appeal to the ontology's resilience.

Effective Realism promises means to develop the most realist-friendly response. There's hope for realism if viewing an effective Lagrangian through the lens of Effective Realism reveals an substantive and resilient ontology. It is time to examine this hope.

3 Trusting effective theories

Now we're going to suppose that our trusty Lagrangian is an effective Lagrangian, and that our reasons for trusting it include, in addition to its capacity to underwrite successful empirical predictions, the fact, crucial to the version of Effective Realism I consider here, that it's been anointed as "robust" by RG considerations.[8] Let's start with a quick review of what all that means.

An effective Lagrangian $\mathcal{L}^{eff}$ does not purport to describe either phenomena nor reality perfectly accurately at all scales. Rather, it purports to describe phenomena up to and including (which I'll often abbreviate as "at" in what follows) a scale μ tolerably accurately. Think of μ as an energy scale, e.g., the energies accessible in our best particle accelerators. (In particle physics, thanks to the ubiquity of wave phenomena and the inverse relation between energy and wavelength, high energies correspond to short distances, making "ultraviolet" a synonym for "high energy.") Presumably $\mathcal{L}^{eff}$'s tolerable accuracy derives from its capacity to roughly reproduce the implications, for μ-scale phenomena, of a more fundamental theory given by a Lagrangian $\mathcal{L}^{UV}$ that describes phenomena up to a very high (ultraviolet cutoff) energy $\Lambda_{UV} >> \mu$. On this way of understanding effective theories, they are, within their scales of application, able mimics of more fundamental theories.

(Wilsonian) RG analyses articulate mimicry relations. Let $\mathcal{L}^{UV}$ be a high-energy (cutoff scale Λ_{UV}) Lagrangian, and consider an effective Lagrangian that mimics its experimental implications well enough at scale μ. RG analyses describe how the effective Lagrangian changes as the cutoff scale is lowered—as higher energy degrees

[8] The "interpreting effective realism" literature is growing, and there is by no means a consensus among commentators concerning how to articulate and engage the view. For other perspectives, see Dougherty (forthcoming), Rivat (2021) and Williams (2015).

of freedom are, as it were, "integrated out" of the cutoff theory. RG transformations act by modifying the couplings giving a Lagrangian's address in coupling space $\mathcal{G}$: as higher-energy details are integrated out of the high energy theory, the couplings modifying the interactions that remain in the effective Lagrangian change in a way that summarizes the impact the blurred-out details have on low energy physics. This change, in the couplings defining effective Lagrangians as they absorb high energy details, constitutes the *RG flow*. RG analyses equip a space of Lagrangians $\mathcal{G}$ with flows connecting Lagrangians defined at higher energies to their lower-energy descendants, where the principle of heredity is coarse-graining.

A Rough Sketch of Implementation Details.[9] QFTs generate predictions by integrating expressions involving a Lagrangian against a range of energies. $\mathcal{L}^{eff}$ "applies up to scale μ" in the sense that it's integrated against energies up to μ in order to make predictions (say, of observable quantities $\mathcal{M}$) testable at energies up to μ.[10] Schematically: $\mathcal{M} = \int_0^\mu \ldots \mathcal{L}_{eff} \ldots$. Able mimicry requires:

$$\int_0^{\Lambda_{UV}} \ldots \mathcal{L}^{UV} \ldots =_{<\mu} \int_0^{\mu} \ldots \mathcal{L}^{eff} \ldots \qquad \text{[MIMIC]}$$

where " $=_{<\mu}$" denotes agreement about predictions testable at energies up to μ. We can use [MIMIC] to characterize a map $R_{\mu\Lambda_{UV}}$ on the space of Lagrangians: for $\mathcal{L}^{eff}$ and $\mathcal{L}^{UV}$ satisfying [MIMIC], $\mathcal{L}^{eff} = R_{\mu\Lambda_{UV}}\mathcal{L}^{UV}$. Allowing the energies indexing the map to vary generates a family of maps $R_{\Lambda\Lambda'}$ (for $\Lambda < \Lambda'$). When that family of maps satisfies the natural sanity constraint that $R_{\Lambda\Lambda''} = R_{\Lambda\Lambda'}R_{\Lambda'\Lambda''}$ (for $\Lambda < \Lambda' < \Lambda''$), they form a *renormalization group*. Given a high-energy theory ($\mathcal{L}^{UV}$, Λ_{UV}), you can coarse-grain it as follows to obtain an effective scale Λ Lagrangian that mimics it. First, divide the range of integration into two parts, corresponding to energies above and below the new cutoff:

$$\int_0^{\Lambda_{UV}} \ldots \mathcal{L}^{UV} \ldots = \int_\Lambda^{\Lambda_{UV}} \ldots \mathcal{L}^{UV} \ldots \times \int_0^{\Lambda} \ldots \mathcal{L}^{UV} \ldots$$

Now evaluate the integral $\int_\Lambda^{\Lambda_{UV}} \ldots \mathcal{L}^{UV} \ldots$ over the high energy range. This leaves an integral over the lower energy range. Adjust the Lagrangian in its integrand to absorb the consequences of the high-energy modes you just integrated out. The adjusted Lagrangian is $\mathcal{L}^{eff} = R_{\Lambda\Lambda_{UV}}\mathcal{L}^{UV}$, the scale Λ effective Lagrangian. (This "integrating out maneuver" is much harder to pull off in practice than in principle. Polchinski, 1984 illustrates some subtleties and how to navigate them.)

Terminological warning: "RG" is said in many ways, as is "effective theory." (Wilsonian) RG analyses track how the couplings defining an effective Lagrangian change in

[9] See Williams (2023, Sect. 4.3), for a more thorough introduction.

[10] Dougherty (forthcoming) is a discussion, attentive to consequences for effective realism, of how this approach comes off the rails for gauge theories.

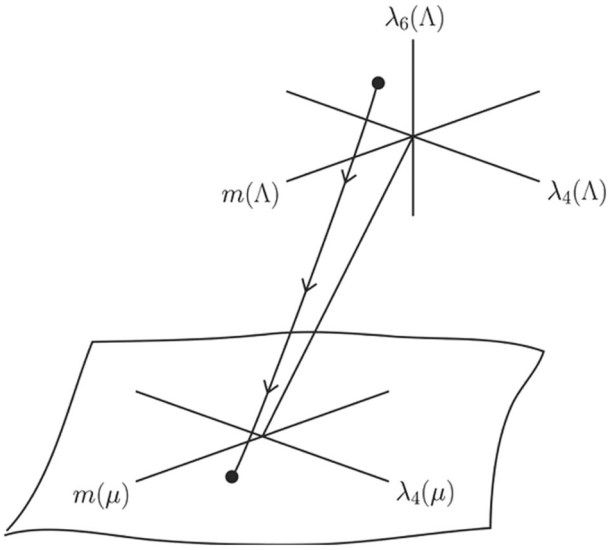

Fig. 3 The RG picture (from Fraser, 2018, Fig. 1). The upper axes represent the infinite-dimensional space of UV-cutoff Lagrangians. (Only three of its axes are visible in the picture.) The upper dot represents an arbitrary element of this space. The arrow emanating from the dot represents the RG flow. The lower dot represents the scale μ Lagrangian to which the RG maps the arbitrary UV-cutoff Lagrangian. The lower dot resides in a *finite*-dimensional space of scale μ Lagrangians, represented by the lower pairs of axes. (Compare Duncan, 2012, Fig. 17.13; Williams, 2023, Fig. 1)

response to *coarse-graining*, to eliminating (usually high energy) degrees of freedom from the theoretical description. A conceptually very different analysis, also (tragically) called "the RG," plays a critical role in the application of interacting QFTs. This Gell-Mann–Low RG approach studies how the couplings defining a QFT must vary with the scale at which renormalization conditions are set, in order for the empirical predictions of that QFT to be insensitive to the choice of renormalization scale. (See Fraser, 2021 for more about both varieties of RG analysis.) Please note that while coarsegraining a high energy theory is one way to obtain an effective theory that is its low energy counterpart, it's not the only way to formulate an effective theory. We'll encounter another strategy, one not obviously mediated by Wilsonian RG analyses, in Sect. 4.

Pivotal to the version of Effective Realism I consider here is an evocative picture, one I'll call the *RG Picture*, of how RG flows organize theory space. Figure 3 presents the RG picture. This section treats the RG picture in soft focus; Sect. 4 will zoom in. The space $G_{\Lambda_{UV}}$ of possible high energy cutoff Lagrangians—represented (incompletely) by the upper trio of axes in the figure—is infinite dimensional. (This is because there are infinitely many distinct interactions that might appear in a Lagrangian governing physics at scale Λ_{UV}. The couplings of those interactions supply a Lagrangian's coordinates in $G_{\Lambda_{UV}}$.) The upper dot represents an arbitrary element of $G_{\Lambda_{UV}}$. The arrow emanating from that dot represents the RG flow mapping UV cutoff scale Lagrangians to the effective Lagrangians they induce, via coarsegraining, at lower energy scales.

The striking thing about the RG Picture is that this flow *carries every Lagrangian* in the infinite dimensional space $G_{\Lambda_{UV}}$ of high energy Lagrangians to a *finite dimensional space* G_μ of scale μ Lagrangians. Coarse-grain an arbitrary Lagrangian in $G_{\Lambda_{UV}}$ enough, and you'll wind up with an Lagrangian in G_μ. Continuing to coarsegrain will simply move the effective Lagrangian around G_μ, but will not change the roster of interactions appearing it. In other words, *no matter what the details of more fundamental physics*, a Lagrangian effective at scale μ depends on the same finite set of interactions. Different effective Lagrangians—Lagrangians induced by different cutoff theories, or by different coarsegrainings of the same cutoff theory—differ only in the couplings that modify those interactions.

And that's not all. The interactions parameterizing G_μ are exactly the interactions that are *renormalizable*. And this isn't just a pipe dream. It's the upshot of Polchinski's celebrated analysis of RG flows on the space of weakly coupled scalar field theories. Pick a UV-cutoff Lagrangian and start the RG flow. At first the news is bad. "Although the Lagrangian might start with a simple form, at lower scales it becomes complicated" (Polchinski 1984, p.277), as interactions of all sorts, not just renormalizable ones, scramble to encapsulate the empirical implications of the high energy modes you've integrated out. But good news is coming:

> At scales far below Λ_{UV} …a great simplification will occur. …no matter what initial Lagrangian we start with (within limits), the lagrangian will be strongly attracted toward a three-dimensional submanifold in the infinite-dimensional space of possible Lagrangians. (1984, p. 277)[11]

According to Polchinski's analysis, non-renormalizable interactions are ineluctably suppressed by the RG flow toward the infrared. Lagrangians in G_μ, a finite-dimensional low-energy surface of attraction for the RG flow, are built only from renormalizable interactions.

This enables the RG Picture to resolve a puzzle Sect. 2.3 posed. The puzzle was: why should we be so lucky that the Lagrangians that describe the physics of our world fall into the tiny corner of theory space containing renormalizable Lagrangians? Casting those Lagrangians as Lagrangians effective at experimentally accessible scales μ, the RG Picture answers: because the RG flow carries low energy effective Lagrangians to a surface spanned by renormalizable interactions. "In light of the above RG analysis," Williams remarks, "it no longer seems like incredible luck that so many useful QFT [Lagrangians] are renormalizable" (2023, p. 54; compare Fraser, 2020a, p. 410).

Effective Realism deploys the RG picture to articulate a version of realism resistant to the Pessimistic Meta-Induction. A notorious anti-realist argument based in the history of science, the Pessimistic Meta-induction argues that our current favored theories T_{now} are likely to be replaced by successor theories that make us want to re-examine (put more bluntly, reject) ontological commitments guided by T_{now}. What Psillos has termed "divide and conquer" responses announce a selection criterion that divides T_{now}'s contents into those that do genuine representational work (and

[11] "within limits" flags a complication my presentation glosses over. It's that Polchinski's moral presupposes cutoff theories reside near the origin of coupling space. See Wallace (2019) for further discussion.

are therefore likely to survive future theory change), and idle theoretical wheels. The selective realist conquers the Pessimistic Meta-Induction by endorsing only the former.

Effective Realism is a strategy for selective realism. Implicating the RG approach to effective theories in a divide and conquer strategy (2019, p. 218), Porter Williams contends that "the RG identifies 'robust' structures in [Effective Field Theories], thereby offering guidance about which structures …represent physical content, and which are just mathematical artifacts" (2019, p. 220). In a similar vein, having recounted the Polchinski analysis depicted in the RG Picture, James Fraser comments that "these [RG] results give us the means to develop a selective realist reading of current QFTs" (2018, p. 1171).

The variety of Effective Realism I consider here takes the RG picture to explicate a sense (or senses) in which an effective Lagrangian can be *robust*. The RG picture assures us that no matter what the cutoff energy truth is, and no matter how finely or coarsely its implications for μ-scale physics are specified, effective physics at scale μ will be given by a Lagrangian that depends only on the finite set of renormalizable interactions parameterizing G_μ. Realist commitments guided by those interactions will be robust in the sense that nothing we learn about actual high energy physics will prevent those interactions from adequately describing scale μ physics. Such commitments will also be robust in something like a safety sense: varying the higher energy truth won't disrupt the capacity, of the ontology committed to, to convey effective physics.

Section 2.4 challenged to realists to read resilient ontologies off of QFT Lagrangians. Effective Realism rises to meet the challenge: RG-informed robustness considerations not only tell us how to read an ontology off of an effective Lagrangian, but also constitute reasons to rely on the resilience of that ontology.

But what ontology, exactly, do robustness considerations enable us to identify on behalf of an effective Lagrangian? In the aftermath of initial, compelling, but programmatic presentations of Effective Realism, no consensus (of which I'm aware) has emerged. Effective Realism's leading progenitors acknowledge this. Here's Fraser:

> In order to substantiate the claim that effective field theories are capturing unobservable aspects of the world, …we need a precise characterization of the non-fundamental descriptive content of QFT models. This is a difficult problem. (Fraser, 2020b, p. 289)[12]

For Williams, Effective Realism's "fundamental obstacle" is that "it is simply unclear that there are any satisfactory candidates for ontological commitment" (2023, p. 70).

In the absence of an explicit consensus Effective Realist ontology, there is a natural ontological default to consider: a folk ontology pegged to the effective Lagrangians Effective Realism celebrates as robust. Here I'll adopt the working hypothesis that the ontology of Effective Realism results from applying folk ontological strategies to effective Lagrangians residing in a low-energy finite dimensional surface of attraction for the RG flow. It's important to keep in view the possibility that Effective Realists can

[12] Fraser (2020b) offers a positive proposal—low energy correlation functions—and defends it against the criticism, offered by Ruetsche (2020) (and developed by Rivat, 2021), that correlation functions live too near the empirical surface of QFTs to serve as a locus of distinctively realist commitment. See Williams (2023, pp. 70–71), for an amicus brief in support of the defense.

deflect the criticisms I offer here by rejecting this ontological hypothesis and offering a better one in its stead! (This is also a good time to emphasize that Effective Realism is still under development; that the version I consider here is a composite sketch of an interesting and cohesive view I, at least, glimpse in its programmatic presentations; and that the actual views of each of Effective Realism's actual advocates could be considerably more resourceful than the version I discuss here.)

That's Effective Realism's central idea. I've presented it as an idea about commitments similar to those constituting the folk ontology, enhanced by an account of what stabilizes those commitments against disruptions arising from unknown physics—their anointment as robust by RG considerations. The problem, I'm about to suggest, is that the enhancement intensifies the difficulty of reading resilient ontologies off of QFT Lagrangians, and does so in a way that undercuts the case for Effective Realism.

4 Blurrier and blurrier

Section 2.4 suggested that the journey, from a successful Lagrangian to a resilient ontology, by way of folk ontological strategies is precarious. Might thinking of the Lagrangian as an effective Lagrangian, in the embrace of distinctive reasons for trusting effective theories, secure safe passage? Alas, I'll try to argue here, the maneuver *amplifies* the sorts of ontological illegibility that matter to questions of realism.

4.1 Prima facie tensions

At the outset, let's note a tension between Effective Realism's account of a Lagrangian worth trusting and a pillar of the folk ontology. That pillar is the thought that *particle species are individuated by their nomically significant traits*—their charge and mass and spin, etc. Lending the thought an air of sanctity is the catechism, associated with Wigner, that elementary particles *are* representations, labelled by spins and masses, of the Poincaré group. An extreme version of the thought is that what it is to be (say) an electron is to have spin $\frac{1}{2}$, mass .511 MeV (and charge -1).

RG approaches wreak havoc with this thought, and in several ways. One is that they could harbor heresies against Wigner's catechism. Natural ways to make sense of the cutoffs specifying an effective theory's domain of application disrupt that theory's Poincaré symmetry. It's often suggested (e.g., Duncan, 2012, p. 370) that this disruption has a happy consequence for those trying to make sense of perturbative QFT. Haag's theorem shows that a dominant framework for perturbatively calculating empirical quantities on behalf of an interacting QFT, a framework known as the interaction picture, is consistent only in the absence of interactions (see Earman & Fraser, 2006; Miller, 2018 for more). The happy consequence is this: because Haag's theorem assumes Poincaré symmetry, when it comes to effective theories, it has no bite. But such a defanging of Haag's theorem comes at a cost: if the Poincaré group isn't a symmetry group of an effective Lagrangian $\mathcal{L}^{eff}$, why should representations of the Poincaré group have anything to do with $\mathcal{L}^{eff}$'s particle content?

RG approaches mount a second affront to the idea that particle species are individuated by their nomically significant traits. Charges and masses are couplings, and in RG approaches couplings "run": they change as the cutoff scale is lowered, a change it is the prime directive of RG approaches to characterize. Running, couplings can't serve as essential attributes individuating particle species. But if one limits the attributes individuating particle species to ones that don't run, one can no longer distinguish between particle species the folk ontology counts as separate. Consider the up, charmed, and top quarks. The attribute distinguishing between them is mass. But masses run. Indeed, the central aim of Grand Unified Theories is to show that masses and interaction couplings, run to high enough energies, converge, marking the merger of forces and particles.[13]

It might be doing the folk ontology a favor to relieve it of the foregoing account of particle species individuation. Declaring it so relieved, let's turn to other tensions between Effective Realism's account of why we should trust an effective Lagrangian and the aspiration to read a resilient ontology off of such a Lagrangian.

4.2 Terms written in invisible ink

Underlying some of these tensions is a complication to the story Sect. 3 told. Section 3 presented the RG picture to first approximation. The approximation omitted something potentially of ontological importance. It's that the *exact* image, of a UV cutoff Lagrangian under the RG flow to scale μ, will typically depend, not only on the finite collection of renormalizable interactions parameterizing G_μ, but also on *every interaction parameterizing the space of cutoff Lagrangians*. It's just that coefficients of non-renormalizable interactions get suppressed—shrunk by factors of $(\frac{\mu}{\Lambda_{UV}})^n$—by the action of the RG (Polchinski, 1984; see Duncan, 2012, Ch. 17, for elaboration). Effective at scale μ Lagrangians with these suppressed interactions written in invisible ink (look like they) reside in a finite dimensional surface G_μ. Fully expressed—that is, encapsulating the implications the RG extracts from the cutoff theory in full specificity— effective at scale μ Lagrangians reside (just as we assumed cutoff Lagrangians to reside) in an infinite dimensional surface.

"The" effective Lagrangian, the one written in invisible ink, is an approximation *even to effective physics* at the scale μ. Written in invisible ink, our trusty effective Lagrangian may look like it deploys only the fields ϕ_1 and ϕ_2, say, and involves them only in a small handful of interactions. But this appearance could be an artefact of an approximation.

What if the terms written in invisible ink make an ontological difference? Then there's a question: which ontology—the one based on $\mathcal{L}^{eff} \in G_\mu$ or the one based on an effective Lagrangian with invisible ink terms restored—is "the" effective ontology? If it's not clear how to answer this question, effective Lagrangians aren't ontologically legible. And I don't think it's at all clear how to answer the question.

Suppose we attribute $\mathcal{L}^{eff}$ an ontology based on *every* interaction, including the infinitely many non-renormalizable ones, parameterizing the infinite-dimensional space $\mathcal{G}$ of Lagrangians. Here's a preliminary and impressionistic account of why

[13] I'm grateful to Alex Blum for this angle.

Effective Realists might not welcome this ontological strategy. To configure $\mathcal{G}$ is to generate a list of interactions furnishing $\mathcal{G}$'s axes. To configure $\mathcal{G}$ is to set the stage for RG analyses of Lagrangians residing in $\mathcal{G}$. But it is not to conduct that analysis. To conduct that analysis requires articulating how trajectories defined by coarse-graining pierce the space. This articulation could reveal those trajectories to organize the space as the RG picture depicts, into one sustaining low energy surfaces of attraction for the RG flow. Thus the *configuration* of $\mathcal{G}$ is less thoroughly informed by the detailed consideration of QFTs and how they take on empirical content than the *RG analysis* of $\mathcal{G}$. In this sense, ontological policies keyed solely to the configuration of $\mathcal{G}$ are less *naturalist*—less sensitive to the specifics of physics and how we use it—than ontological policies keyed to $\mathcal{G}$'s RG analysis. And the ontological policy at issue—attribute $\mathcal{L}^{eff}$ an ontology based on *every* interaction parameterizing $\mathcal{G}$—is keyed solely to the configuration of $\mathcal{G}$. Contrary to Effective Realists' assertion that the details of RG flows help realists to identify robust commitments, the policy severs the content of effective ontology from the *details* of *how* RG flows organize the space $\mathcal{G}$ of Lagrangians. The policy attaches the same ontology to Lagrangians in $\mathcal{G}$, whether or not they're subject to RG analysis, and (if they are) no matter how the analysis goes. Whereas Effective Realism promised a naturalistic and selective strategy for realism duly informed by the details of RG analysis, this ontological strategy appears to deliver none of those things. This is prima facie reason Effective Realists may not welcome it.

If, on the other hand, we base our ontology only on interactions written in *visible* ink, we need to justify drawing an ontological distinction of kind on the basis of an empirical distinction of degree. And it's not obvious how to do that without undermining the project of naturalistic metaphysics. The *austere* naturalistic metaphysician, who withholds commitment from empirically "insignificant" ontological posits, may turn out to be no metaphysican at all. There's a long history of practicing physicists who abjure ontology but wrest successful empirical predictions from effective Lagrangians. This suggests that Lagrangians have empirical content whether we assign them ontologies or not. If the suggestion holds water, it leaves the austere naturalistic metaphysician without an ontology to commit to!

The foregoing difficulties are ones Sect. 2.3 anticipated: while we may have a tolerable grasp on what is for one quantity to approximate another, what it is for one *ontology* to approximate another is a far more elusive matter. Indeed, on the basis of a sustained analysis of neglected and important differences between empirical and metaphysical approximation, McKenzie concludes that "approximation seems inapplicable to canonical claims of metaphysics" (2020, p. 7; see also McKenzie, 2024). In light of such challenges, we should take seriously the possibility that terms written in invisible ink compromise $\mathcal{L}^{eff}$'s ontological legibility, *even if those terms are empirically negligible*. To make matters worse, we've been too quick to assume that those terms are negligible.

We've been assuming that for practical (if not for metaphysical!) purposes, we can get away with proceeding as follows. First, take the exact image of a cutoff theory under the RG flow to scale μ—an effective Lagrangian incorporating infinitely many interactions. Second, write all its exponentially suppressed terms in invisible ink to obtain something that looks like an element of the finite-dimensional low energy attractive surface G_μ. Third, calculate with that "invisible ink" Lagrangian. And maybe we can

get away with that procedure. Until we can't. Contributions to empirical quantities can be negligible in the sense of *small* without being negligible in the sense of *bereft of physical import*. Tony Duncan makes the point vividly. Suppressed ("irrelevant," in Duncan's vernacular) interactions can contribute to low-probability processes "sufficiently different …from the processes induced by relevant interactions to stand out phenomenologically" (2012, p. 576). He continues

> the dimension-six four-fermion [interaction] of Fermi weak interaction theory, responsible for β decay, for example, is 'irrelevant' from this point of view, but the associated vast phenomenology of radioactivity is hardly so. (2012, p. 576)

The example is worth elaborating.[14] In one sort of β decay, a neutron in the nucleus of an atom decays into a proton, changing the atom's atomic number and releasing a bunch of energy in the process. Not every substance suffers β decay—the ones that do are *radioactive*. Radioactive substances don't suffer β decay very often. The half-life of Uranium^{238}—the waiting period such that quantum mechanics assigns probability $\frac{1}{2}$ to the eventuality that an atom of Uranium^{238} β decays to Thorium^{234} during that period —is upwards of 4 billion years!

Fermi's "4 Fermi" theory was an early theory of β-decay. (Very early—Fermi proposed it only a year and a half after the discovery of the neutron!) The 4 Fermi Lagrangian includes an interaction term involving 4 fermions: a neutron, a proton, an electron, and an anti-neutrino. Figure 1 exhibited a first order Feynman diagram for that interaction. A standard commentary on that figure dresses the 4 Fermi theory in its folk ontology. In β-decay, a neutron (n) decays into a proton (p), releasing two decay products: a high energy electron (e^-) and an electron antineutrino ($\bar{\nu}_e$). A "very successful model of radioactive decay" (Schwartz, 2014, p. 396), the 4 Fermi theory, applied in lower energy regimes, accurately describes relations between the rates of different β-decay processes, as well as their angular dependence and energy distributions.

But there's a catch: the interaction in the 4 Fermi Lagrangian responsible for Fig. 1 is non-renormalizable! According to the RG picture, non-renormalizable interactions are suppressed at low energies, and thereby rendered unworthy of parameterizing a low energy finite-dimensional surface of attraction for the RG flow. Suppressed interactions are the ones the RG picture enjoins us to write in invisible ink. In the context of the 4 Fermi theory, the low probability of radioactive decay reflects the exponential suppression of its non-renormalizable interactions. But it also illustrates the maxim that low probability events can abound with physical significance. It follows from this maxim that writing an effective Lagrangian's suppressed terms in invisible ink can efface its physical significance. Put another way, there can be significant ontological differences—differences as profound as the difference between worlds with and without radioactivity—between an effective Lagrangian's visible and invisible ink ontologies.

An ontology pegged to an effective Lagrangian whose suppressed terms are written in invisible ink can be inadequate *even to phenomena at effective scales the Lagrangian is meant to address*. But writing its suppressed terms in visible ink plucks an effective

[14] See Calderón, (ms) for more extensive discussion of this case and its consequences for Effective Realism.

Lagrangian from the finite-dimensional low energy surface of attraction that's the linchpin of Effective Realist arguments for a distinctive variety of robustness exhibited by effective theories in the ambit of RG analyses. Ontologies informed by invisible ink terms might not be ontologies RG considerations protect against the Pessimistic Meta-Induction.

Another aspect of this example merits remark. Moderns regard the 4 Fermi theory as the effective theory implementing the Electroweak ($SU(2) \times U(1)$) sector of the Standard Model at low energies. But this regard is not based on a Wilsonian RG analysis of the sort that frames Fig. 3's RG Picture. That is, the 4 Fermi theory is not obtained by coarse-graining the Electroweak Lagrangian. Rather, "matching conditions" are invoked to identify the 4 Fermi theory as an effective theory that conveys the low-energy consequences of the Electroweak theory. Very roughly, the invocation proceeds as follows: Work at an energy scale μ "low" in the sense that it's small compared to the energy m_W required to create the heavier particles of the Electroweak theory. Use the Electroweak Lagrangian to calculate amplitudes for processes at these energies. Then look for a Lagrangian—any Lagrangian; don't sweat whether or not it's renormalizable—that doesn't use the heavier fields but does output low energy amplitudes roughly matching those of the Electroweak theory. The 4 Fermi theory fits the bill. Although this can be described as "integrating heavier particles out" of the higher energy theory to obtain the effective theory, the identification typically proceeds *not* by the derivation of coarse-grained consequences but by comparing the outputs of perturbative approximations guided by the Lagrangians in question. (For details see Schwartz, 2014, Sect. 31.3.1; Georgi, 1993).

Effective theories so obtained sit awkwardly within the RG Picture.

> in the effective theory, we replace the nonlocal interactions from virtual heavy-particle exchange with a set of local interactions, constructed to give the same physics at low energies …The result of eliminating heavy particles is inevitably a *nonrenormalizable* theory, in which the *nontrivial effects* of the heavy particles appear in [nonrenormalizable] interactions with dimensions higher than 4. (Georgi, 1993, 212)

The awkwardness is that the RG picture exhorts us to write non-renormalizable interactions in invisible ink—but these are the very interactions the matching strategy relies on to construct effective Lagrangians!

The Chiral Lagrangian describing short range nuclear forces (see Schwartz, 2014, Sect. 22.3)—the effective Lagrangian reckoned to encapsulate the low-energy content of QCD—suffers a similar awkwardness. Connected to QCD not by a derivation that "integrates out" QCD's higher energy details, but by exceedingly indirect empirical matching conditions, the Chiral Lagrangian makes use of non-renormalizable interactions. It also deploys a roster of fields from which fields appearing in the QCD Lagrangian are absent. This shows that when an effective Lagrangian is obtained by matching, it's not just the couplings that can change to incorporate the impact of omitted physics, but also the roster of interactions and even fields appearing in the Lagrangian. Such changes carry the effective Lagrangian outside a low-energy surface of attraction, parameterized only by renormalizable interactions, depicted by the RG Picture.

The Chiral Lagrangian and the 4 Fermi Lagrangian are exemplars of effective theories in use in high energy physics. Neither of them conforms to the RG Picture undergirding Effective Realism. Their non-conformity mars the beautiful explanation the RG picture offered of the apparent happy accident that the QFTs governing the actual world just happen to be renormalizable. The QFTs that govern the regimes of actual world where we conduct experiments, that explanation went, are low-energy effective QFTs; the RG picture shows that such QFTs are confined to G_μ, a finite-dimensional low energy surface of attraction for the RG flow, a surface spanned by renormalizable interactions; hence those theories are renormalizable. What mars the beautiful explanation is the circumstance, witnessed by the 4 Fermi theory and the chiral Lagrangian, that significant effective theories aren't always renormalizable! They're exiles from the surface G_μ that serves Effective Realism as a platform from which to construct a resilient effective ontology. For those who would be realists about these exemplary effective theories, Effective Realism offers no guidance.

4.3 Symmetry challenges

The previous section addressed impediments to resiliently ontologizing an effective Lagrangian arising from the circumstance that effective Lagrangians, including elements of a finite dimensional surface of attraction G_μ central to the RG picture, are themselves approximate. A resident of G_μ is an "invisible ink" approximation to a more detailed Lagrangian, typically one depending on infinitely many interactions, that one would obtain were one to trace the RG flow from the UV in all specificity. But when it comes to effective Lagrangians, another sort of approximation is afoot. Itself approximate, an effective Lagrangian is also a springboard for approximation techniques. These can be applied to a Lagrangian without prejudice to whether it's regarded as effective or not. But regarding it as effective—as applicable to low energy phenomena—warrants the pursuit of approximation schemes adapted to low energy applications.

Ruiz de Olano et al. (2022, Sect. 4) develop a decidedly more nuanced and concrete illustration of the moral that effective Lagrangians aren't ontologically legible in isolation. Their case study of the Nambu–Jona–Lasinio model analyses that model as "a new approximation method fixed to an old Hamiltonian" —the Heisenberg Hamiltonian—in a way that "clarif[ies] its particle content" (93). Viewed, as it were, with the naked eye, the Heisenberg Hamiltonian appeared incapable of describing massive fermions. Viewed through the lens of the Nambu–Jona–Lasinio approximation scheme, a method that exploits the spontaneous breaking of chiral symmetry, the Heisenberg Hamiltonian's physical content looks different: "the achievement of the new approximation scheme was to demonstrate that a model with exact chiral symmetry could describe massive fermionic particles" (82). This supports Ruiz de Ruiz de Olano et al's *Content Determination Thesis*, that appoximations can play a pivotal role in assigning physical content. When they do, the theoretical core in question—the Heisenberg Hamiltonian of Ruiz de Olano et al's careful case study and the effective Lagrangian at issue here—fails on its own to specify an ontology.

There is another way symmetry considerations can complicate effective ontology.[15] Sometimes, symmetries are our best guide to the form of an effective Lagrangian—only their guidance falls short of fixing that form. Consider the question of how to identify an effective Lagrangian implementing QCD at low energies. Matching procedures that unfold order by order in perturbation theory helped to identify the 4-Fermi theory as a low-energy effective theory implementing the Electroweak theory. But matching procedures of this sort are no help here. Because QCD's couplings grow as energy decreases, it can't be treated perturbatively in low-energy regimes. In practice, physicists follow a different route from QCD to an effective Lagrangian. The QCD Lagrangian has an approximate chiral symmetry: if we pretend quarks have 0 mass, it's invariant under interchange of left- and right-handed states. Moreover, that symmetry is thought to be spontaneously broken in low-energy states.[16] The alternate route is to "forget all about QCD and just use the symmetry breaking pattern to write down an effective description" (Schwartz, 2014, p. 568). That is, construct a chirally symmetric low-energy Lagrangian, and explore the consequences of spontaneously breaking that chiral symmetry.

The catch is that the recipe underdetermines the effective Lagrangian (see Weinberg, 1967 for a classic discussion). To see this, we need only consider the first step of the recipe—devising a low-energy Lagrangian that's chirally symmetric. The linear sigma model is one way to realize the symmetries guiding the effective description. The effective Lagrangian for the linear sigma model involves nucleon and pion fields, as well as a scalar meson field whose presence is required to respect chiral symmetry. It's renormalizable. The non-linear sigma model is another way to realize the symmetries guiding the effective description. The scalar field can be eliminated from the non-linear sigma model's effective Lagrangian without loss of chiral symmetry. That effective Lagrangian incorporates nucleon and pion fields only—but includes non-renormalizable interactions. Prominent among these is a pion self-interaction term that can involve infinitely many "soft" pions—pions with energies so low that there can be infinitely many of them without violating energy conservation. The linear and the non-linear sigma models underwrite effective Lagrangians that invite very different ontologies: very different particle ledgers and very different accounts of how those particles interact. Both are prima facie candidate effective ontologies. Of course, much more can be said in the service of adjudicating their merits. But enough has been said to indicate that the symmetry considerations that constrain an effective Lagrangian on their own fall short of fixing an effective ontology. This is another (defeasible!) sense in which the turn to effective theories exacerbates the ontological legibilty of QFT Lagrangians.

[15] I am obliged to an anonymous referee for pointing me toward this example.

[16] See Schwartz (2014, Ch. 28), for an overview of spontaneous symmetry breaking. But it will suffice for present purposes to treat the notion as a placeholder that mediates between the QCD Lagrangian and its effective counterparts.

5 Refocus: humble empiricism

I openly and cheerfully admit that the foregoing considerations are consistent with many morals and reactions other than the ones I've tried to extract and express. Still, I want to close by describing something the considerations suggest to me (even if they don't prove it to you). They suggest (to me!) that "the" form of the Lagrangian, especially if it's an approximate effective Lagrangian $\mathcal{L}^{eff}$, is malleable; viewing $\mathcal{L}^{eff}$ through the lenses of different intended applications/approximation techniques/theoretical agendas/etc can yield ontologies that differ importantly from one another. Legitimate reactions to this include attempting to adduce grounds to privilege a particular ontology, or attempting to motivate approaches to ontology under which the apparent differences melt away, and the evidently rival ontologies blend into one. Should either attempt succeed, we'd be left in the enviable position of knowing what ontology to believe, if we trusted $\mathcal{L}^{eff}$. And in that sense we'd understand physics based on $\mathcal{L}^{eff}$.

But, I don't think we should dissolve into lamentation if attempts to attach a "winning ontology" to $\mathcal{L}^{eff}$ fail. One way to understand a piece of physics is to find a winning ontology for it—an ontology that makes sense of the wide-ranging and variegated successes that got us, qua ontologists, interested in that piece of physics to begin with, an ontology that will prove resilient in face of the future-directed march of physics. But there's no guarantee that we're going to pull that off. And in cases where we don't, another sort of understanding may be available. I call the view, that science should aim to produce understanding of this more modest sort, *humble empiricism*.

Consider an effective Lagrangian $\mathcal{L}^{eff}$ that's, all things considered, ontologically illegible. $\mathcal{L}^{eff}$ and the uses to which it is put stymie our best attempts to identify a winning and resilient ontology on its behalf. Instead, $\mathcal{L}^{eff}$ suggests a collection of ontologies, not all of them mutually consistent, and none of which triumphs overall. There could nevertheless be local and limited domains where a specific ontology is a winner. (And there may be a version of Effective Realism keyed to some of those ways!) For instance, taking ghost fields in $\mathcal{L}^{eff}$ seriously might win for purposes internal to calculation, such as regularization or gauge fixing. Taking goldstone bosons seriously might win when applying $\mathcal{L}^{eff}$ at low energy. And reconceptualizing a lowly parameter appearing in $\mathcal{L}^{eff}$—for instance, QCD's θ angle— as itself a field associated with novel particles —axions—could win for the purposes of addressing foundational problems—the strong CP problem— and charting successor theories.

In cases where no ontology for $\mathcal{L}^{eff}$ triumphs overall, we have an alternate route to (and sense of) understanding $\mathcal{L}^{eff}$. It's to have a handle on what range of ontologies it admits, how it manages its admissions process (paying special attention to instances of the Content Determination Thesis!), and how those ontologies fare in different sorts of competitions. Perhaps this is metaphysically unsatisfying. But it stands to be physically fruitful. $\mathcal{L}^{eff}$ is, after all, not final, but effective physics. Entertaining a range of ways to apply and conceptualize $\mathcal{L}^{eff}$, we entertain as well a range of ways forward (a resource celebrated in Sklar, 2000, Ch. 4). And that stands to be a superior strategy for creative science than narrow adherence to a single ontology we've declared the winner in order to sustain a stringent sense of understanding that might stand at cross-purposes with working physics.

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